

Beyond the conjectured bound

Sharp discrepancy for a family of binary substitutions

Counterexamples and corrected theorems for
OEIS A284365 and A284366

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Abstract

The OEIS entries A284365 and A284366 conjecture that two complementary position sequences associated with the substitution $0 \mapsto 1, 1 \mapsto 101010$ have linear-approximation errors smaller than 2. Both upper bounds are false. Their first counterexamples occur at ranks 8 933 313 and 2 977 771, respectively, and have the same error 2.0048721733261288 We give exact integer certificates for these values and for their minimality. More substantially, we determine the complete discrepancy limit sets for every substitution $0 \mapsto 1, 1 \mapsto (10)^m$, $m \geq 1$. For the OEIS case $m = 3$, the sharp common upper bound is $(6 + \sqrt{21})/5 = 2.116515138991168 \dots$. The proof uses two state-dependent prefix-weight intervals and a contracting set recursion. We also prove the companion density conjectures, construct explicit infinite families of counterexamples, derive rational generating functions for their indices, and provide exact-arithmetic random-access and verification software. The mathematical proofs are self-contained; finite experiments are supporting checks, not substitutes for the proofs.

Status and scope. The cited OEIS pages still labeled the relevant statements as conjectures when accessed on September 19, 2026. This article establishes the stated results by explicit proofs and executed certificates. It does not claim an exhaustive priority search, peer review, or proof-assistant formalization. General spectral methods for substitution discrepancy are classical; the contribution here is the exact solution and certificates for this family and these particular conjectures.

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1 The selected problem and its resolution

Let σ be the substitution

$$\sigma(0) = 1, \quad \sigma(1) = 101010. \quad (1.1)$$

Its fixed word beginning with 1 is

$$\mathbf{w} = 101010110101011010101 \cdots.$$

This is A284364 [1]. Number positions starting at 1. Let $u(n)$ and $v(n)$ be the positions of the n th zero and the n th one, respectively. Thus u is A284365 and v is A284366 [2, 3]. Put

$$r = \frac{9 + \sqrt{21}}{6}, \quad s = \frac{\sqrt{21} - 1}{2}, \quad \frac{1}{r} + \frac{1}{s} = 1. \quad (1.2)$$

The position-sequence entries conjecture

$$-2 < rn - u(n) < 2, \quad -1 < sn - v(n) < 2 \quad (n \geq 1). \quad (1.3)$$

They also conjecture their respective densities to be $(9 - \sqrt{21})/10$ and $(1 + \sqrt{21})/10$. The density comments are dated April 13, 2025; the sequence entries credit Clark Kimberling, March 26, 2017. We use the unambiguous position-sequence formulations: the parent entry's comment uses count terminology alongside position slopes, which would be a different and immediately incompatible statement.

Theorem 1.1 (Resolution for the OEIS sequences). *Write $E_0(n) = rn - u(n)$ and $E_1(n) = sn - v(n)$. For every $n \geq 1$,*

$$-\frac{12 + 2\sqrt{21}}{15} < E_0(n) < \frac{6 + \sqrt{21}}{5}, \quad (1.4)$$

$$\frac{\sqrt{21} - 9}{5} < E_1(n) < \frac{6 + \sqrt{21}}{5}. \quad (1.5)$$

Each endpoint is a subsequential limit, and indeed the full set of subsequential limits of each error sequence is the corresponding closed interval. In particular, both conjectured lower bounds in (1.3) hold, but both upper bounds fail infinitely often.

The first counterexamples are

$$v(2\,977\,771) = 5\,334\,043, \quad u(8\,933\,313) = 20\,222\,898. \quad (1.6)$$

Their common error is

$$\frac{2\,977\,771\sqrt{21} - 13\,645\,857}{2} = 2.00487217332612884794005732278 \dots > 2. \quad (1.7)$$

The two asserted densities are correct.

The bounds are not consequences of extrapolating the observed maxima. Sections 2–5 prove them for an entire family. Section 6 supplies a finite certificate for (1.6) and its minimality. Section 7 gives an explicit sequence of errors increasing to the sharp upper endpoint. The method belongs to the established study of the relation between substitution spectra and discrepancy [4]; all specific interval and sharpness arguments needed here are proved below.

2 A parameterized family and its discrepancy theorem

Fix an integer $m \geq 1$ and define

$$\sigma_m(0) = 1, \quad \sigma_m(1) = (10)^m. \quad (2.1)$$

The image of 1 begins with 1 and has length at least 2, so the words $\sigma_m^k(1)$ form nested prefixes of an infinite fixed word $\mathbf{w}^{(m)}$. The substitution contains both letters in an iterate of either letter; no additional initial condition is needed beyond the initial 1.

Throughout the general argument let

$$q = \frac{\sqrt{m^2 + 4m} - m}{2}, \quad \lambda = m + q = \frac{m}{q}, \quad s = 1 + q, \quad r = 1 + \frac{1}{q}, \quad (2.2)$$

and

$$t = 1 - q, \quad C = \frac{q}{1 - q^2}, \quad H = qC = \frac{q^2}{1 - q^2}. \quad (2.3)$$

The quantities depend on m , even when the subscript is suppressed. The following identities will be used repeatedly:

$$0 < q < 1, \quad q^2 = m(1 - q), \quad q(1 + H) = C, \quad H(1 + q) = m, \quad \frac{1}{r} + \frac{1}{s} = 1. \quad (2.4)$$

For example, $H(1 + q) = q^2/(1 - q) = m$. Also q is irrational: $(m + 1)^2 < m^2 + 4m < (m + 2)^2$ for $m \geq 1$.

Let $Z(N)$ and $O(N)$ count the zeros and ones in the first N letters, with $Z(0) = O(0) = 0$. Let $u_m(n)$ and $v_m(n)$ denote the corresponding position sequences, and set

$$E_0(n) = rn - u_m(n), \quad E_1(n) = sn - v_m(n).$$

Theorem 2.1 (Sharp error intervals for every m). *For every integer $m \geq 1$ and $n \geq 1$,*

$$C - m < E_1(n) < C, \quad (2.5)$$

$$\frac{1 - C}{q} < E_0(n) < C. \quad (2.6)$$

The closures of the two error-value sets are exactly these intervals with their endpoints included. These are also their full subsequential limit sets. No endpoint is attained at a finite index.

For every $N \geq 0$,

$$-\frac{H}{1 + q} < O(N) - \frac{N}{1 + q} < \frac{C}{1 + q}. \quad (2.7)$$

Again, the two endpoints are sharp, and the closure of the centered counting errors is the full closed interval. In particular,

$$\lim_{N \rightarrow \infty} \frac{O(N)}{N} = \frac{1}{s}, \quad \lim_{N \rightarrow \infty} \frac{Z(N)}{N} = \frac{1}{r}.$$

A useful equivalent form of the lower zero-position endpoint is

$$\frac{1 - C}{q} = C - 1 - \frac{m - 1}{q}. \quad (2.8)$$

For $m = 3$, substituting $q = (\sqrt{21} - 3)/2$ gives

$$C = \frac{6 + \sqrt{21}}{5}, \quad H = \frac{3 + 3\sqrt{21}}{10},$$

and (2.5)–(2.6) become (1.5)–(1.4).

3 A contracting weight and two prefix states

For a finite binary word P , define its weight by

$$X(P) = |P|_0 - q|P|_1. \quad (3.1)$$

This is additive under concatenation. The incidence matrix of the substitution, with row order 0, 1 and column order 0, 1, is

$$M = \begin{pmatrix} 0 & m \\ 1 & m \end{pmatrix}.$$

Its eigenvalues are λ and $-q$. More directly, the letter weights give the identity on which the proof rests:

Lemma 3.1 (Weight contraction). *For every finite word P ,*

$$X(\sigma_m(P)) = -qX(P). \quad (3.2)$$

Proof. For $P = 0$ the two sides equal $-q$. For $P = 1$, the image has m letters of each type and hence weight $m(1 - q) = q^2 = -q(-q)$. Additivity proves the claim for an arbitrary concatenation of letters. $\square$

The next letter is crucial. For $a \in \{0, 1\}$, let

$$S_a = \{X(w_1 \cdots w_N) : N \geq 0, w_{N+1} = a\}. \quad (3.3)$$

Thus $0 \in S_1$, from the empty prefix. A prefix ending immediately before a zero belongs to state 0, not to a state determined by its last letter.

Lemma 3.2 (Exact state equations). *The sets in (3.3) satisfy*

$$S_0 = \bigcup_{j=0}^{m-1} (-qS_1 + jt - q), \quad (3.4)$$

$$S_1 = (-qS_0) \cup \bigcup_{j=0}^{m-1} (-qS_1 + jt). \quad (3.5)$$

Proof. View the fixed word as the concatenation of images of its own letters. If a parent prefix P is followed by 0, its image is followed by the single letter 1, giving the transition $-qS_0$ in (3.5). If P is followed by 1, its image is followed by $(10)^m$. Immediately before the $(j + 1)$ st one in that image, the internal prefix is $(10)^j$, whose weight is jt . Immediately before the corresponding zero, the internal prefix is $(10)^j1$, whose weight is $jt - q$. Lemma 3.1 gives all the stated affine images. Conversely, every location in the fixed word lies in one of these image blocks. Thus there are no missing transitions and the equations are set equalities. $\square$

Using a single state and adding all possible offsets independently would forget which parent letter permits which offset. The two-state system retains precisely the information required for the sharp constants.

4 Solving the interval recursion, including sharpness

Define two closed intervals

$$I_0 = [-C, H - 1], \quad I_1 = [-qH, H]. \quad (4.1)$$

They are nondegenerate for every $m \geq 1$.

Lemma 4.1 (Invariant intervals). *The pair (I_0, I_1) satisfies (3.4)–(3.5) with S_a replaced by I_a .*

Proof. Put $J_j = -qI_1 + jt$. Each J_j has length

$$qH(1 + q) = mq,$$

while successive left endpoints are separated by $t = 1 - q = q^2/m$. If $m \geq 2$, then $mq \geq t$, so adjacent intervals overlap. For $m = 1$ there is only one interval. The extreme endpoints yield

$$\bigcup_{j=0}^{m-1} J_j = [-qH, H - t]. \quad (4.2)$$

Indeed, the upper endpoint is $q^2H + (m - 1)t = H - t$, since $q^2(H + 1) = H$ and $mt = q^2$.

Subtracting q from (4.2) gives

$$\bigcup_{j=0}^{m-1} (J_j - q) = [-qH - q, H - t - q] = [-C, H - 1] = I_0.$$

For state 1, the other image is

$$-qI_0 = [q - qH, H].$$

It overlaps or touches the interval in (4.2), because

$$H - t \geq q - qH \iff H(1 + q) \geq 1 \iff m \geq 1.$$

Their union is therefore $[-qH, H] = I_1$. □

Lemma 4.2 (Strict inclusion of all actual prefixes). *Every element of S_a lies in the interior of I_a .*

Proof. We induct on the length N of the prefix. The empty prefix has weight 0, which lies strictly inside I_1 .

For a nonempty prefix, locate its next letter in an image block. The associated parent prefix has some length $k < N$. To see strict decrease, if $k > 0$ then that parent prefix begins with 1, so its image has length at least $k + (2m - 1) > k$; this image is contained in the original prefix. If $k = 0$, strict decrease follows from $N > 0$.

By induction, the parent weight belongs to the interior of its appropriate interval. Its actual transition is an affine map with nonzero slope $-q$. The image of an interior point is an interior point of the image interval, and Lemma 4.1 puts that image interval inside the appropriate target interval. Such a point cannot be an endpoint of the target interval. This completes the induction. □

Bounds alone do not prove sharpness. We next show that the intervals contain no gap in the closures of the actual prefix weights.

Lemma 4.3 (Uniqueness for the compact set recursion). *Let K_0, K_1 be nonempty compact subsets of $\mathbb{R}$ satisfying the state equations. Then $K_0 = I_0$ and $K_1 = I_1$.*

Proof. For compact nonempty subsets of $\mathbb{R}$, write d_H for Hausdorff distance. An affine map of slope $-q$ multiplies this distance by q . Also, for finite indexed families,

$$d_H\left(\bigcup_i A_i, \bigcup_i B_i\right) \leq \max_i d_H(A_i, B_i).$$

Let $D = \max\{d_H(K_0, I_0), d_H(K_1, I_1)\}$. Apply these observations to the two state equations for K and I . They imply $D \leq qD$. Since D is finite and $0 < q < 1$, it follows that $D = 0$. $\square$

Proposition 4.4 (Exact closures).

$$\text{cl } S_0 = [-C, H - 1], \quad \text{cl } S_1 = [-qH, H]. \quad (4.3)$$

The closure of all prefix weights is $[-C, H]$.

Proof. Lemma 4.2 makes S_0, S_1 bounded, and both are nonempty. Take closures in their exact finite-union equations. Closure commutes with each affine map and with a finite union, so $(\text{cl } S_0, \text{cl } S_1)$ is a compact fixed pair. Lemma 4.3 proves (4.3). Finally I_0 and I_1 overlap or touch, since $H - 1 \geq -qH$ is again equivalent to $m \geq 1$. Their union is $[-C, H]$. $\square$

Distinct prefix lengths have distinct weights. Indeed, equality would give $\Delta Z = q\Delta O$, impossible for irrational q unless both integer differences vanish, in which case the lengths agree. A set dense in a nondegenerate interval has infinitely many distinct points in every relative neighborhood of every point of that interval. Consequently its density statements remain valid after deleting any finite number of prefixes. This proves that the closures just obtained are also the full sets of limits along prefix lengths tending to infinity.

5 Position errors, densities, and exact transfer

5.1 Passing from prefix weights to position errors

Immediately before the n th one, the prefix has $n - 1$ ones and $v_m(n) - n$ zeros. Its weight is therefore

$$X = v_m(n) - n - q(n - 1) = q - E_1(n). \quad (5.1)$$

Immediately before the n th zero, the prefix has $n - 1$ zeros and $u_m(n) - n$ ones, giving

$$X = n - 1 - q(u_m(n) - n) = qE_0(n) - 1. \quad (5.2)$$

Apply the strict state bounds and Proposition 4.4. Equation (5.1) transforms I_1 into

$$[q - H, q + qH] = [C - m, C],$$

and (5.2) transforms I_0 into $[(1 - C)/q, C]$. The identities follow from (2.4). The state sets contain exactly the prefixes immediately before the respective ranked letters, so no additional selection restriction is needed. This proves the position part of Theorem 2.1, including strictness and the full subsequential limit sets.

For ordinary prefix counts,

$$X = N - (1 + q)O(N), \quad O(N) - \frac{N}{1 + q} = -\frac{X}{1 + q}. \quad (5.3)$$

The total prefix-weight interval $[-C, H]$ proves (2.7) and its sharpness. Dividing the bounded counting error by N establishes the densities without invoking an ergodic theorem. In the case $m = 3$, this yields the quantitative estimate

$$-\frac{33 + 3\sqrt{21}}{50} < O(N) - \frac{1 + \sqrt{21}}{10}N < \frac{27 + 7\sqrt{21}}{50}. \quad (5.4)$$

Thus the density conjectures in all three relevant OEIS entries follow, and the error in the unnormalized counting function is uniformly bounded.

5.2 A relation between the two position sequences

Proposition 5.1 (Exact position transfer). *For $n \geq 1$ and $1 \leq h \leq m$,*

$$u_m(m(n-1) + h) = v_m(n) + (2m-1)n - 2m + 2h. \quad (5.5)$$

Equivalently,

$$E_0(m(n-1) + h) = E_1(n) - (m-h)(r-2). \quad (5.6)$$

In particular,

$$u_m(mn) = v_m(n) + (2m-1)n, \quad E_0(mn) = E_1(n). \quad (5.7)$$

Proof. Before the parent word's n th one there are $n-1$ ones and $v_m(n) - n$ zeros. The total length of their images is $2m(n-1) + v_m(n) - n$. The image of the next one contains m zeros, at internal positions $2, 4, \dots, 2m$, and no zero comes from the image of a parent zero. Thus the h th such zero has global rank $m(n-1) + h$ and the position in (5.5).

Subtract this expression from $r(m(n-1) + h)$. The identity $m(r-2) = q$ reduces the result to (5.6). $\square$

For $m = 3$, $C < r$; explicitly

$$r - C = \frac{9 - \sqrt{21}}{30} > 0.$$

Since $r - 2 > 0$, (5.6) shows that the errors at zero ranks $3(n-1) + 1$ and $3(n-1) + 2$ are strictly less than $C - (r-2) < 2$. It follows that

$$\{k : E_0(k) > 2\} = \{3n : E_1(n) > 2\}. \quad (5.8)$$

Equality to 2 is impossible, because r and s are irrational. This bijection will make the zero-position minimality result immediate once the first one-position violation has been certified.

6 An exact finite certificate and the first counterexamples

6.1 Recursive blocks

Define auxiliary finite words

$$W_{-1} = 0, \quad W_0 = 1, \quad W_k = (W_{k-1}W_{k-2})^m \quad (k \geq 1). \quad (6.1)$$

For $k \geq 0$, induction gives $W_k = \sigma_m^k(1)$. In particular, W_k is a prefix of $\mathbf{w}^{(m)}$. Write L_k, Z_k, O_k for its length, zero count, and one count. Each satisfies

$$a_k = m(a_{k-1} + a_{k-2}). \quad (6.2)$$

For the remainder of this section $m = 3$.

Consider the finite word

$$P = W_{11} W_{10} W_{10} W_8 W_6 W_4 W_2. \quad (6.3)$$

This is a prefix of W_{12} , followed by a 1. Here is an explicit way to follow the recursive decomposition. In $W_{12} = (W_{11}W_{10})^3$, take the first $W_{11}W_{10}$, then enter the next $W_{11} = (W_{10}W_9)^3$ and take its first W_{10} . The residual word begins with W_9 . In $W_9 = (W_8W_7)^3$, take W_8 , leaving a word beginning with W_7 . Repeat with W_7, W_5, W_3 , taking W_6, W_4, W_2 . The remaining word begins with $W_1 = 101010$, so its next letter is 1.

The count recurrence gives

$$|P|_0 = 2\,356\,272, \quad |P|_1 = 2\,977\,770, \quad |P| = 5\,334\,042. \quad (6.4)$$

Consequently the next one proves the first identity in (1.6). Formula (5.7) proves the second:

$$u(3 \cdot 2\,977\,771) = 5\,334\,043 + 5 \cdot 2\,977\,771 = 20\,222\,898.$$

Their common error is exactly (1.7). To certify that it exceeds 2, it suffices to observe

$$21(2\,977\,771)^2 - (13\,645\,861)^2 = 265\,940 > 0. \quad (6.5)$$

Both quantities compared before squaring are positive, so (6.5) proves $2\,977\,771\sqrt{21} > 13\,645\,861$ with no numerical approximation.

6.2 Why no earlier index is a counterexample

For any finite word U , define

$$\mu_1(U) = \min\{X(Q) : Q \text{ is a proper prefix of } U, \text{ and the next letter in } U \text{ is } 1\}.$$

Use $+\infty$ if U has no one. For concatenations,

$$\mu_1(UV) = \min\{\mu_1(U), X(U) + \mu_1(V)\}, \quad (6.6)$$

$$\mu_1(B^m) = \mu_1(B) + \min\{0, (m-1)X(B)\}. \quad (6.7)$$

The second formula follows by considering the m translated copies of B . These identities give a small exact certificate for all earlier positions at once.

Apply (6.6) to the seven factors of P . The translated candidate minima, in their order of appearance, are

$$\begin{aligned} &431\,712 - 545\,583q, & 1\,820\,700 - 2\,300\,934q, \\ &2\,222\,379 - 2\,808\,561q, & 2\,346\,957 - 2\,965\,998q, \\ &2\,355\,624 - 2\,976\,951q, & 2\,356\,227 - 2\,977\,713q, \\ &2\,356\,269 - 2\,977\,766q. \end{aligned} \quad (6.8)$$

Their minimum is the first, so

$$\mu_1(P) = 431\,712 - 545\,583q. \quad (6.9)$$

All of these finite comparisons can be checked by the integer sign rule in Section 8; the separate checker in the archive recomputes the block table and its minimum rather than accepting them as input.

k	L_k	Z_k	O_k	$\mu_1(W_k)$
0	1	0	1	0
1	6	3	3	0
2	21	9	12	$6 - 8q$
3	81	36	45	$9 - 12q$
4	306	135	171	$99 - 126q$
5	1 161	513	648	$144 - 183q$
6	4 401	1 944	2 457	$1 440 - 1 821q$
7	16 686	7 371	9 315	$2 088 - 2 640q$
8	63 261	27 945	35 316	$20 718 - 26 184q$
9	239 841	105 948	133 893	$30 033 - 37 956q$
10	909 306	401 679	507 627	$297 819 - 376 374q$
11	3 447 441	1 522 881	1 924 560	$431 712 - 545 583q$
12	13 070 241	5 773 680	7 296 561	$4 280 832 - 5 409 957q$

Table 1: Block data for $m = 3$. Every minimum is calculated by (6.6)–(6.7), using exact quadratic-field comparisons. The $k = 12$ row is included for context; only the preceding rows are needed to certify P .

By (5.1), the largest one-position error strictly before the candidate equals

$$q - \mu_1(P) = 545\,584q - 431\,712 = 1.988978393507077166 \dots < 2. \quad (6.10)$$

The last inequality has the exact certificate

$$(2\,500\,180)^2 - 21(545\,584)^2 = 110\,224 > 0. \quad (6.11)$$

It proves $545\,584\sqrt{21} < 2\,500\,180$, which is precisely (6.10) < 2 after substituting the value of q . The maximum in (6.10) occurs at rank 545 584, at position 977 296.

Every earlier ranked one lies in P . Hence no earlier one-position error is at least 2. Equations (6.5) and (6.11) certify the first counterexample, and the exact bijection (5.8) certifies the first zero-position counterexample. This proof covers all earlier indices; it is not a claim based only on a finite sample of selected positions.

7 Infinite extremal families and their generating functions

The compact-set proof gives sharpness nonconstructively in terms of closures. A specific family approaching the common upper endpoint can also be written down and evaluated by short recurrences.

Proposition 7.1 (Explicit extremal prefixes). *For general $m \geq 1$, let P_0 be the empty word and define*

$$P_{j+1} = \sigma_m^2(P_j)(10)^m, \quad U_j = \sigma_m(P_j). \quad (7.1)$$

Every P_j and every U_j is a prefix of the fixed word, followed by a 1. Set

$$n_j = |U_j|_1 + 1, \quad p_j = |U_j| + 1. \quad (7.2)$$

Then

$$v_m(n_j) = p_j, \quad E_1(n_j) = C(1 - q^{2j+2}) \nearrow C. \quad (7.3)$$

The zero-position errors at ranks mn_j are equal to these errors.

Proof. Suppose P_j is a prefix followed by 1. Its image is followed by $(10)^m$. Therefore $\sigma_m(P_j)1$ is a prefix followed by 0. Apply the substitution again: the resulting prefix is $\sigma_m^2(P_j)(10)^m = P_{j+1}$, followed by $\sigma_m(0) = 1$. This proves the prefix assertion inductively from P_0 . It also shows that U_j is followed by 1.

By Lemma 3.1 and $m(1 - q) = q^2$,

$$X(P_{j+1}) = q^2 X(P_j) + q^2.$$

Starting at zero and summing the geometric progression gives $X(P_j) = H(1 - q^{2j})$. Thus

$$E_1(n_j) = q - X(U_j) = q + qH(1 - q^{2j}) = C - Cq^{2j+2},$$

where $q + qH = C$ and $qH = Cq^2$. The errors increase strictly because $0 < q < 1$. The zero-position assertion follows from (5.7). $\square$

Let $(z_j, o_j) = (|P_j|_0, |P_j|_1)$. The preceding construction is fully integer-recursive:

$$\begin{pmatrix} z_{j+1} \\ o_{j+1} \end{pmatrix} = \begin{pmatrix} m & m^2 \\ m & m^2 + m \end{pmatrix} \begin{pmatrix} z_j \\ o_j \end{pmatrix} + \begin{pmatrix} m \\ m \end{pmatrix}, \quad (z_0, o_0) = (0, 0), \quad (7.4)$$

with

$$n_j = z_j + mo_j + 1, \quad p_j = z_j + 2mo_j + 1. \quad (7.5)$$

Eliminating the two counts yields an especially simple result.

Corollary 7.2 (Scalar recurrences and rational generating functions). *Put $T = m(m + 2)$. Then*

$$n_{j+2} = Tn_{j+1} - m^2n_j + 1 - m, \quad n_0 = 1, \quad n_1 = m^2 + m + 1, \quad (7.6)$$

$$p_{j+2} = Tp_{j+1} - m^2p_j + m^2 - m + 1, \quad p_0 = 1, \quad p_1 = 2m^2 + m + 1. \quad (7.7)$$

Their ordinary generating functions are

$$\sum_{j \geq 0} n_j z^j = \frac{1 - mz}{(1 - z)(1 - m(m + 2)z + m^2 z^2)}, \quad (7.8)$$

$$\sum_{j \geq 0} p_j z^j = \frac{1 + m(m - 1)z}{(1 - z)(1 - m(m + 2)z + m^2 z^2)}. \quad (7.9)$$

Proof. The linear part in (7.4) is M^2 , with trace T and determinant m^2 . Its characteristic identity implies the two inhomogeneous recurrences after substitution in (7.5); alternatively, one may eliminate z_j, o_j directly. The constant terms and initial values follow by evaluating $j = 0, 1, 2$.

For a sequence obeying $a_{j+2} = Ta_{j+1} - m^2a_j + c$, multiplying its series by $1 - Tz + m^2z^2$ gives $a_0 + (a_1 - Ta_0)z + cz^2/(1 - z)$. For (7.6), the numerator after multiplication by $1 - z$ reduces to $1 - mz$; for (7.7), it reduces to $1 + m(m - 1)z$. This proves both generating functions. $\square$

In the OEIS case the recurrences are $n_{j+2} = 15n_{j+1} - 9n_j - 2$ and $p_{j+2} = 15p_{j+1} - 9p_j + 7$. Formula (7.3) shows that every $j \geq 6$ gives a counterexample. The closed-interval theorem is stronger still: every open interval contained in $(2, C)$ contains infinitely many errors of each position sequence, not just those in this special family.

j	n_j	$p_j = v(n_j)$	$E_1(n_j)$, rounded
0	1	1	0.791287847478
1	13	22	1.286742017213
2	184	328	1.596963935937
3	2 641	4 729	1.791205189187
4	37 957	67 990	1.912826719410
5	545 584	977 296	1.988978393507
6	7 842 145	14 047 537	2.036659732964
7	112 721 917	201 917 398	2.066514757944
8	1 620 249 448	2 902 333 144	2.085208077525

Table 2: An explicit extremal family for $m = 3$. It supplies infinitely many counterexamples, but its first violation ($j = 6$) is not the first violation over all ranks.

7.1 Exact asymptotics along the extremal family

The denominator in (7.8)–(7.9) factors as

$$(1 - z)(1 - \lambda^2 z)(1 - q^2 z).$$

Here $\lambda > 1$ and $0 < q < 1$, so partial fractions give an exact three-term expansion. For either sequence $a_j = n_j$ or $a_j = p_j$, let

$$d = \frac{c}{1 - 2m}, \quad A = \frac{a_1 - d - q^2(a_0 - d)}{\lambda^2 - q^2}, \quad B = a_0 - d - A,$$

where c is the corresponding constant in its recurrence. Then

$$a_j = A\lambda^{2j} + d + Bq^{2j}. \quad (7.10)$$

The leading coefficient is positive, as follows either from these expressions or the positive count recurrence. In particular, $n_j, p_j = \Theta(\lambda^{2j})$, whereas

$$C - E_1(n_j) = Cq^{2j+2} = \Theta(p_j^{-\kappa}), \quad \kappa = -\frac{\log q}{\log \lambda}. \quad (7.11)$$

For $m = 3$, $\kappa = 0.17565278149 \dots$. This is a statement along the constructed subsequence, not a claim about the optimal record-error convergence rate over every prefix. The comparatively small exponent explains why substantial growth in the rank can coexist with a slow approach to the limiting extremum.

8 Exact algorithms and executed checks

8.1 Quadratic arithmetic without rounding decisions

Every prefix weight and every one-position error is of the form $a + bq$ with integer a, b . Since $q^2 = m - mq$, addition and multiplication stay in this representation. To compare $a + bq$ with zero, put

$$A = 2a - mb, \quad D = m^2 + 4m.$$

Its sign is the sign of $A + b\sqrt{D}$. If A and b have the same nonzero sign, that sign is the answer. Zero cases are immediate. If they have opposite signs, the answer is

$$\text{sgn}(A) \text{sgn}(A^2 - Db^2). \quad (8.1)$$

As D is nonsquare, there is no unhandled nontrivial equality case. Thus all block-extremum decisions and counterexample inequalities in the software reduce to integer arithmetic. Zero-position errors may contain $1/q$; multiplying their comparisons by the positive number q puts them in the same representation.

8.2 Associative summaries and random access

A block summary stores its zero and one counts, and for each following letter state the minimum and maximum prefix weight, together with a location attaining the extremum. Under concatenation UV , the extrema from V are translated by $X(U)$ and their positions by $|U|$; the appropriate minimum or maximum is then taken with the extrema from U . This is the same identity as (6.6) in both states and both directions.

Precompute summaries for $W_{-1}, W_0, \dots, W_k$ using (6.1). To count the first N letters, find the first block long enough, skip whole child pairs by their lengths, and descend into the residual child. To find the n th specified letter, use its block counts instead of lengths. At a fixed m , there are $O(\log N)$ levels, and each level involves only a bounded number of arithmetic operations. The library groups repeated child pairs for these operations.

To find the first error at least a threshold B , a block beginning after a prefix of weight x can be discarded exactly when its appropriate extremum cannot reach the threshold. For ones the feasibility condition is

$$x + \mu_1(W_k) \leq q - B.$$

For zeros, using the maximum prefix weight $\nu_0(W_k)$ before a zero, the condition is

$$x + \nu_0(W_k) \geq qB - 1.$$

Find a first feasible enclosing block, then inspect children from left to right and descend into the first feasible child, skipping the others by their counts. The returned leaf is the first violation, and the skipped block extrema certify that no earlier one was missed.

For fixed m and a counterexample at a position of size N , the precomputation and descent use $O(\log N)$ arithmetic operations on $O(\log N)$ -bit integers. This is an arithmetic-operation count, not a unit-cost bit-complexity assertion. With $M(b)$ denoting the cost of multiplying b -bit integers, the sign tests give the conservative bound $O(\log N M(\log N))$ bit operations, alongside additions and small-parameter divisions. Storing all summaries uses $O((\log N)^2)$ bits at fixed m . Searching only up to a prescribed level and finding nothing is a finite negative result, not a proof of nonexistence in the infinite word.

8.3 What was actually checked

The delivered script code `verify.py` -long completed successfully under Python 3.13.5. It recorded 310 816 explicit checks, in addition to the assertions internal to the separate certificate checker. For every $m = 1, \dots, 8$, it built a literal 12 000-letter prefix, checked the sharp bounds at every letter, and compared selected random-access queries and complete small-block extrema against direct calculations. It also checked the transfer identity, the exact extremal family through $j = 24$, and the proposed recurrences and generating-function numerators.

For $m = 3$, the script compared the first 30 displayed terms of each position sequence and the first 30 binary letters with the OEIS entries. It did not download or compare the complete linked 10 000-term tables. The optional long test independently formed the fixed-word prefix of length 20 222 898 by literal substitution, confirming both specified counterexample positions and ranks. The first-counterexample search used exact summaries, not a floating-point scan.

The file code/`independent_check.py` does not import the main library. It reconstructs the count and minimum certificate with a separate integer sign test and concatenation implementation. This checks the finite computations supporting minimality; the prefix-membership argument and transfer theorem supply their mathematical justification. The verification data in `data/` records exact coefficients as well as display decimals, and includes the full first-violation descent traces. No experiment is used to infer the infinite interval theorem.

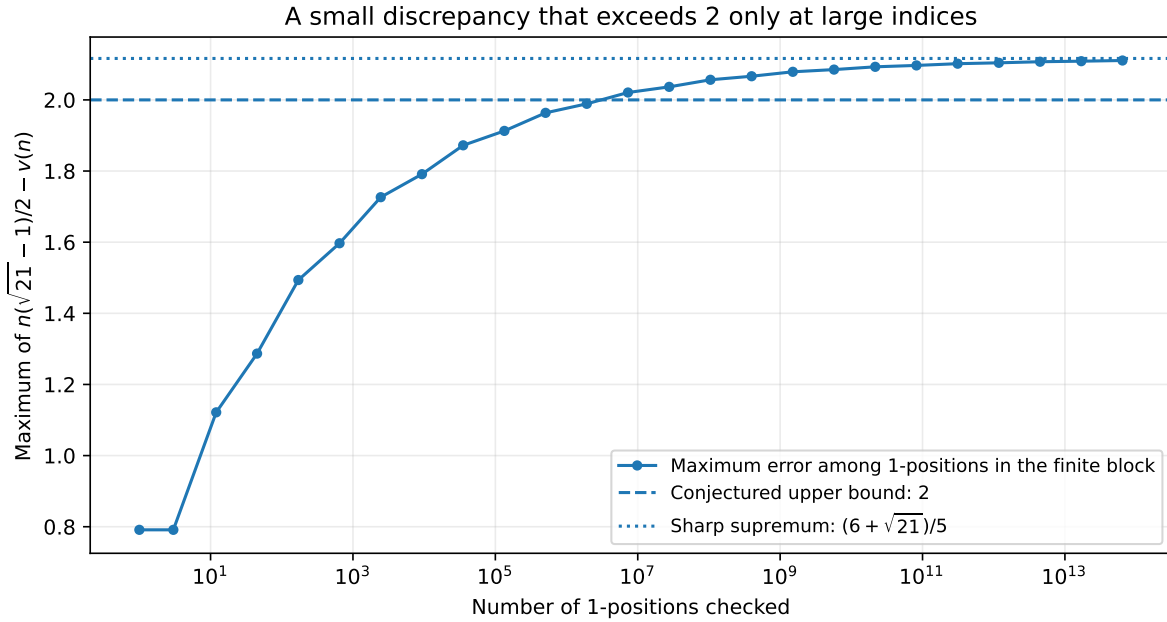


Figure 1: Maximum one-position error over all ones in W_k , for $k = 0, \dots, 24$. The horizontal scale is the number of one-positions in the enclosing block, not the rank at which its maximum is attained. The curve is computed from exact block summaries; floating-point values are used only to render the figure.

9 Further consequences and boundaries of the result

9.1 The $m = 1$ endpoint is a Beatty pair

At $m = 1$, $q = (\sqrt{5} - 1)/2$ and $C = 1$. Both error intervals become $(0, 1)$. Since the positions are integers and the slopes are irrational,

$$v_1(n) = \lfloor (1 + q)n \rfloor, \quad u_1(n) = \left\lfloor \left(1 + \frac{1}{q}\right)n \right\rfloor.$$

Thus the familiar complementary Beatty description is recovered directly from the theorem. For $m \geq 2$, no fixed real shift γ can give $v_m(n) = \lfloor sn + \gamma \rfloor$ for every n : such an expression would put all its errors in an interval of length 1, whereas the error closure in (2.5) has length exactly m . The density already forces any putative Beatty slope to be s .

9.2 A finite factor-balance bound

The centered prefix-counting errors have range width

$$\frac{C + H}{1 + q} = C.$$

For a factor of length L , its number of ones minus L/s is the difference of two prefix errors, so it lies strictly between $-C$ and C . Two factors of the same length consequently differ in their numbers of ones by an integer whose absolute value is strictly less than $2C$. Hence

$$||A|_1 - |B|_1| \leq \lceil 2C \rceil - 1 \quad \text{whenever } |A| = |B|. \quad (9.1)$$

For $m = 3$ this gives a balance bound of 4. The article does not assert that this bound is optimal: sharp prefix extrema do not automatically occur at two endpoints separated by the same prescribed factor length. The exact optimal factor-balance constant is a separate question.

9.3 The word itself has a nonrational generating function

Although the extremal-index series (7.8)–(7.9) are rational, the characteristic series

$$F_m(z) = \sum_{n \geq 1} w_n^{(m)} z^n$$

is not rational. Indeed, a rational power series has coefficients obeying an eventual fixed linear recurrence. If those coefficients take values in a finite set, the vector of a fixed number of consecutive coefficients has only finitely many states and deterministically produces its successor. A state eventually repeats, making the sequence eventually periodic. An eventually periodic binary sequence has rational density, contradicting $1/(1+q)$, which is irrational. This elementary observation concerns the full word, not the rational generating functions of the sparse families.

9.4 What remains outside the claims

The conjectured upper bounds and density statements selected in Section 1 are resolved by the preceding proofs. The article does not determine the frequency distribution of discrepancy values within their interval, the density of the ranks with error above 2, or the exact best factor-balance constant. Knowing the support of subsequential limits does not by itself answer those distributional questions. Nor is the constructed extremal subsequence claimed to enumerate all record errors. These distinctions prevent stronger statements from being inferred merely from the observed finite data or the interval closure result.

10 Conclusion

A conjecture about an error below 2 can survive a substantial finite check while failing only slightly at a much larger index. Here the first one-position violation occurs at rank 2 977 771, and its transfer produces the first zero-position violation at rank 8 933 313. The finite disproof is certified by a short recursive prefix and two integer square comparisons. The structural analysis goes further: the correct common upper bound is $(6 + \sqrt{21})/5$, and a two-state contraction proves exact discrepancy intervals for every $0 \mapsto 1$, $1 \mapsto (10)^m$. Density, explicit extremal families, rational index-generating functions, and efficient exact verification then follow from the same framework.

A Reproducing and inspecting the deliverables

All verification code uses only the Python standard library and requires Python 3.10 or later. From the archive’s top-level directory run:

```
python code/independent_check.py
python code/verify.py --long
```

The shorter command `python code/verify.py` omits the 20-million-letter literal construction but retains the exact first-counterexample search and finite certificate. A successful run writes its own report and regenerates the exact data. Running the shorter command will replace the saved report with one that accurately records that the long test was omitted.

For an individual large-index query:

```
from substitution import Substitution

model = Substitution(3)
assert model.select(1, 2977771) == 5334043
assert model.select(0, 8933313) == 20222898
print(model.first_at_least(1, threshold=2))
```

Run this from `code/`, or add that directory to the Python module search path. Ranks and returned positions are 1-based; prefix lengths and internal offsets are 0-based. The exact search’s `None` result means that no witness was found through the specified maximum block level, not that the infinite sequence has no witness.

The main archive components are described in Table 3. The pre-rendered figure is included, so compiling the article does not require Python or Matplotlib. To rebuild the PDF, run `pdflatex` on `article.tex` twice; the bibliography is embedded in the source. To rebuild only the plot, run `python code/make_figure.py`, which requires Matplotlib. No fonts, checksum files, temporary TeX build files, or downloaded copies of other authors’ papers are included.

File or directory	Purpose
<code>article.tex</code> , <code>article.pdf</code> <code>code/substitution.py</code>	Complete source and compiled article. Exact quadratic arithmetic, summaries, counts, selection, first-threshold search.
<code>code/independent_check.py</code> <code>code/verify.py</code>	Separate compact checker for the finite certificate. Executable cross-checks and data generation.
<code>data/certificates.json</code>	First-counterexample traces and exact certificate coefficients.
<code>data/finite_extrema.json</code>	Block extrema through W_{30} .
<code>data/extremal_family.json</code>	First 25 explicit extremal-family terms.
<code>data/verification.json</code>	Report of the executed test suite.
<code>figures/</code>	The figure in vector PDF and raster PNG form.
<code>oeis_proposed_updates.txt</code>	Suggested mathematical corrections, not submitted to OEIS.
<code>README.md</code>	Build instructions, scope, and verification notes.

Table 3: Archive contents.

References

- [1] The OEIS Foundation Inc., *A284364: Fixed point of the morphism $0 \mapsto 1, 1 \mapsto 101010$* . Entry credited to Clark Kimberling, March 26, 2017; density conjecture comment by Amiram Eldar, April 13, 2025. <https://oeis.org/A284364>. Accessed September 19, 2026.
- [2] The OEIS Foundation Inc., *A284365: Positions of 0 in A284364; complement of A284366*. Entry credited to Clark Kimberling, March 26, 2017; density conjecture comment by Amiram Eldar, April 13, 2025. <https://oeis.org/A284365>. Accessed September 19, 2026.
- [3] The OEIS Foundation Inc., *A284366: Positions of 1 in A284364; complement of A284365*. Entry credited to Clark Kimberling, March 26, 2017; density conjecture comment by Amiram Eldar, April 13, 2025. <https://oeis.org/A284366>. Accessed September 19, 2026.
- [4] B. Adamczewski, *Symbolic discrepancy and self-similar dynamics*, *Annales de l'Institut Fourier* **54** (2004), no. 7, 2201–2234. doi:10.5802/aif.2079. https://www.numdam.org/item/AIF_2004__54_7_2201_0/.